DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
2025-2026 EVEN SEMESTER
MODERN PEDAGOGY VERIFICATION

S. No	Faculty Name with Designation	Date of visit	Class/ Section details	Subject Name	Module details	Pedagogy used
1	Dr. S. Senthilkumar	29.10.2025	II A and B	Control Systems	2	Peer Teaching
2	Dr.M. Malathi	22.04.2026	III A and B	Antenna and Wave Propagation	4	Using lab components
3	Dr.M. Malathi	17.04.2026	III A and B	Antenna and Wave Propagation	4	Real time example
4	Dr.M. Malathi	1.05.2026	III A and B	Antenna and Wave Propagation	3 & 4	Guest lecture
5	Mrs.R.Keerthika	18.03.2026	III A and B	Embedded system design and IoT	4	Hands on session
6	Mrs.K.Vembarasi	27.02.2026	IV A and B	Satellite communication	1	Video demonstration

M. Malathi

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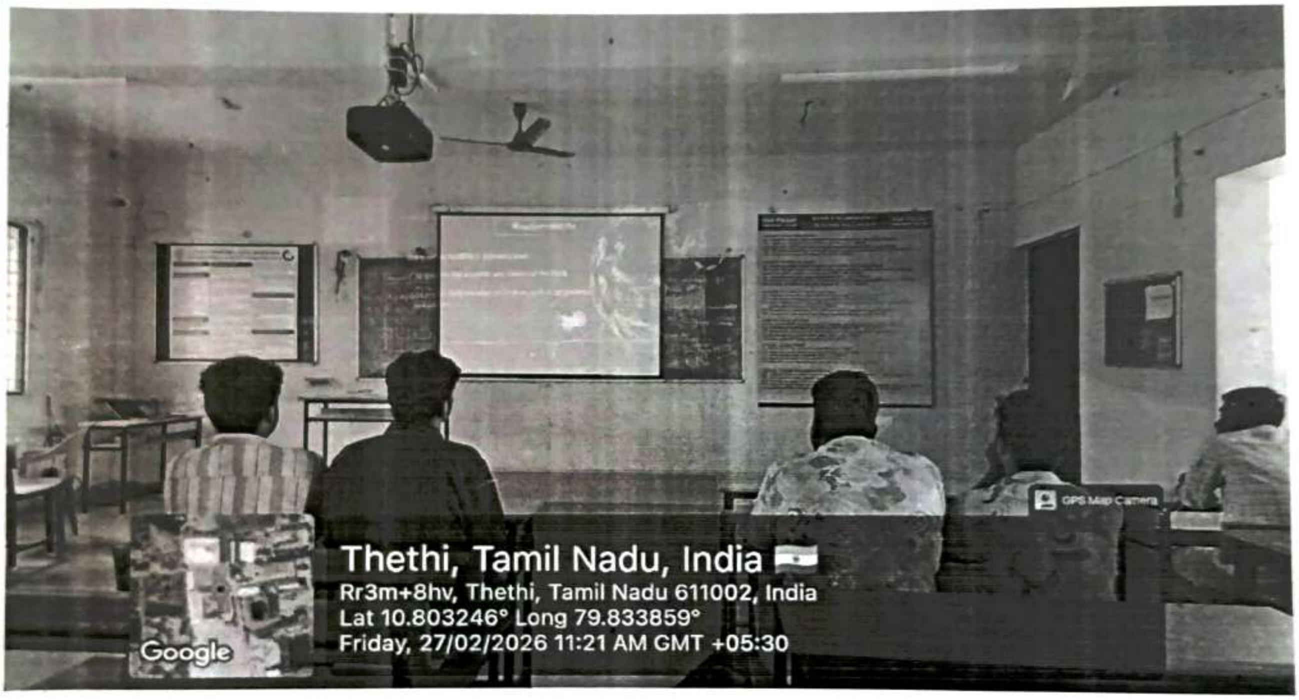
DEPARTMENT OF ELECTRONCIS AND COMMUNICATION ENGINEERING

Academic Year: 2025-26 | Even Semester | 8th Semester B.E Programs | Regulation: 2019

1903EC025 – Satellite communication
Module 1 – satellite launching procedure – 27.2.2026
Video Demonstration



Mrs.K.VEMBARASI, M.E.,(Ph.D),
Assistant Professor
Department of Electronics and

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B.E. – Electronics and Communication Engineering

Academic year: 2025 - 2026 (Even Semester)
Fourth Semester

2302EC403 – Control Systems

Innovative practice

Date: 26.05.2026 **Hour: 2**

Peer Teach

Topic: Controllability and Observability

Based on the students' academic performance and subject understanding, selected students were assigned to conduct a peer teaching session on the topic "Controllability and Observability" in the Control Systems course. The peer tutors explained the fundamental concepts, significance, mathematical conditions, and practical applications of controllability and observability in state-space analysis. They demonstrated the procedures for determining controllability and observability using controllability and observability matrices and solved illustrative numerical problems to enhance understanding. The activity encouraged collaborative learning, improved conceptual clarity among students, and strengthened their problem-solving and communication skills. Active participation from both peer tutors and learners contributed to a better understanding of the topic and enhanced overall classroom engagement.

Outcome of the Activity:

peer teaching session helped students gain a deeper understanding of controllability and observability concepts, improved their confidence in solving state-space analysis problems. After this session, students are solved the following problems.

1. The state space representation of a system is given below.

$$\begin{bmatrix}\dot{X}_1\\ \dot{X}_2\\ \dot{X}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}1\\ 0\\ 1\end{bmatrix}u \text{ and } y = \begin{bmatrix}10 & 5 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Check for controllability and observability

2. Examine the controllability and observability of the system with state equation.

$$\begin{bmatrix}\dot{X}_1\\ \dot{X}_2\\ \dot{X}_3\end{bmatrix} = \begin{bmatrix}0 & 0 & 1\\ -2 & -3 & 0\\ 0 & 2 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 2\\ 0\end{bmatrix}u \text{ and } y = \begin{bmatrix}0 & 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

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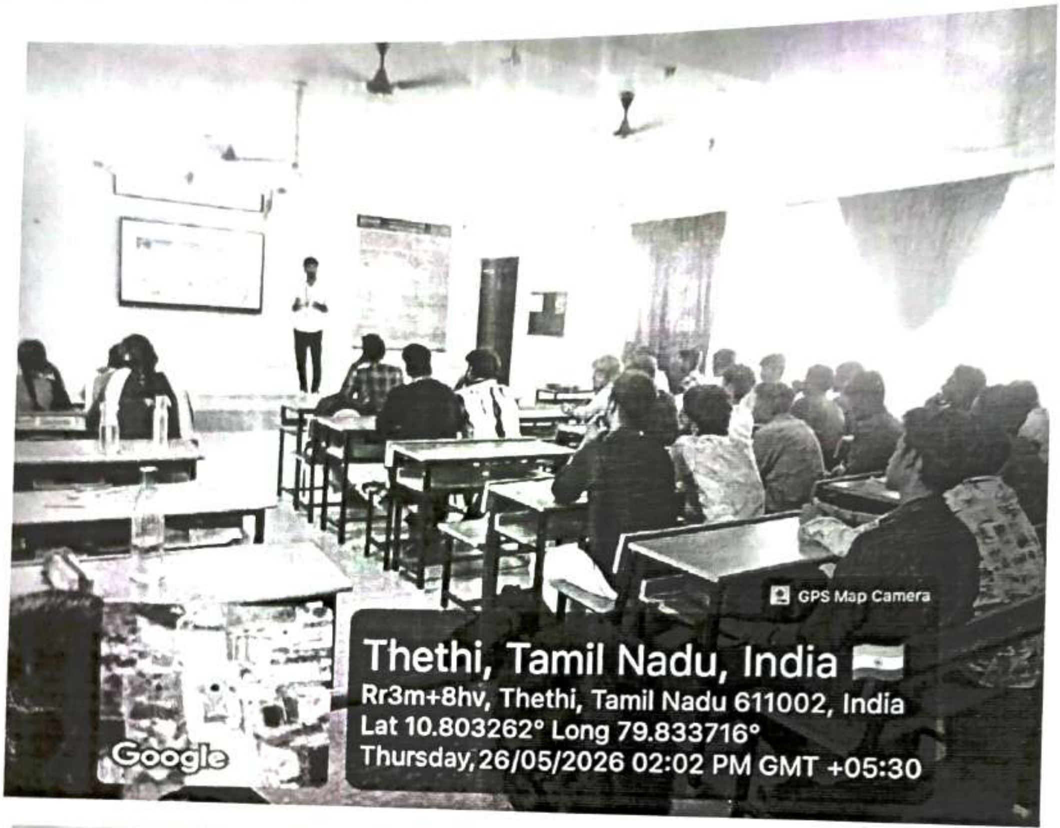

30/05/26
Course Coordinator

Dr.S.SENTHILKUMAR, M.Tech., Ph.D.,
Associate Professor
Department of ECE

30/5
HoD

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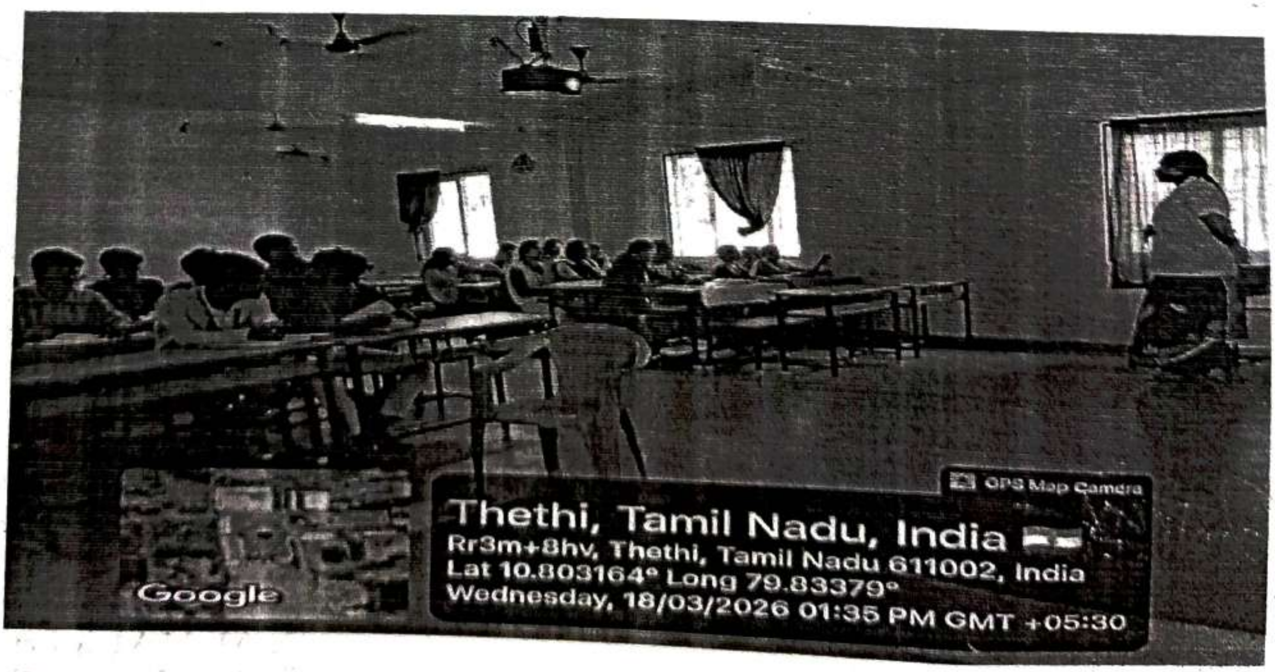
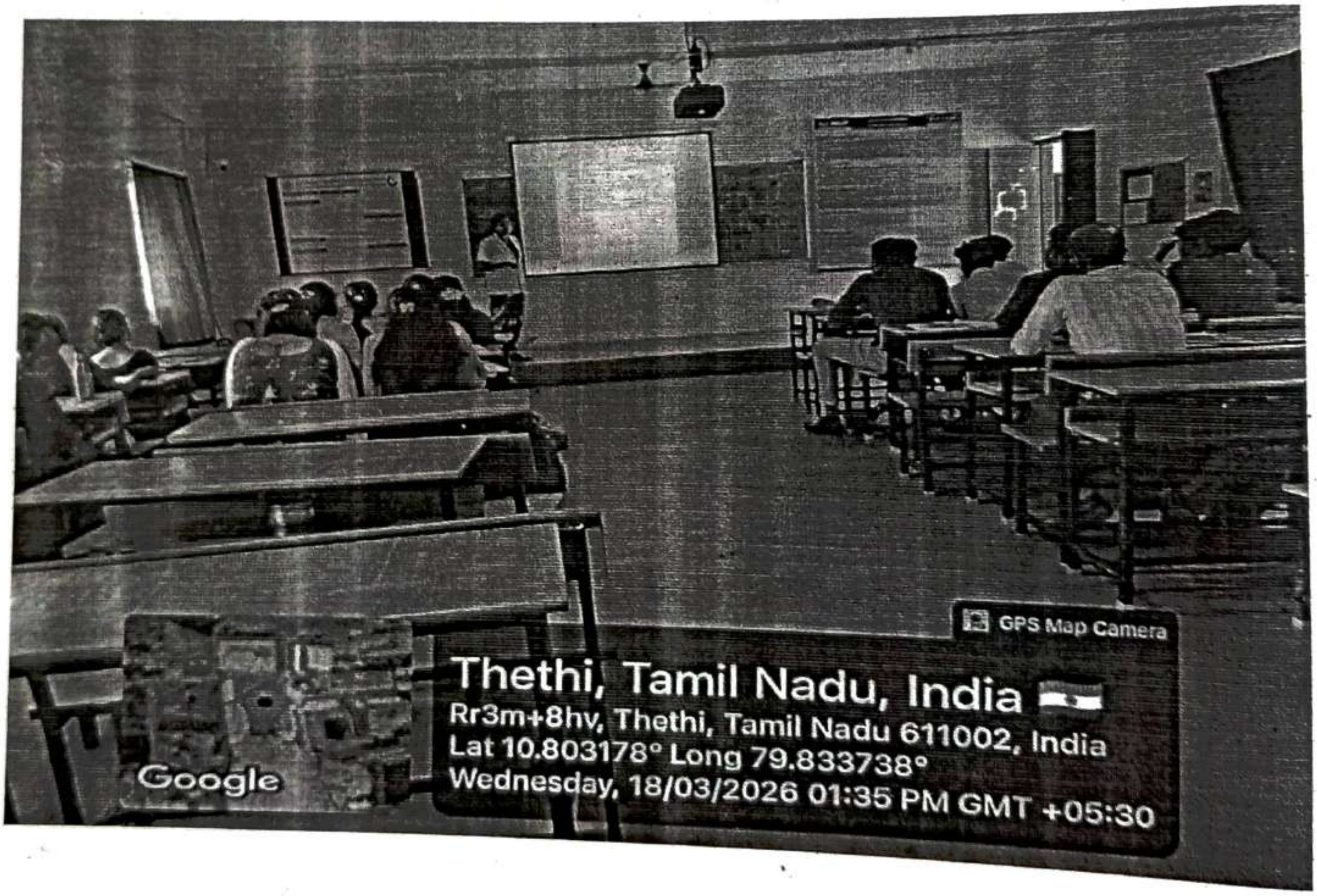
Academic Year: 2025-26 | Even Semester | 6th Semester B.E Programs | Regulation: 20

2302EC602 – Embedded System Design and IOT
Modern Pedagogy in Embedded Systems and IoT Education Using Arduino Uno, ESP32, and Raspberry Pi on 18.03.2026

A technical session on "Modern Pedagogy in Embedded Systems and IoT Education Using Arduino Uno, ESP32, and Raspberry Pi" was conducted on 18 March 2026. The program aimed to enhance students' understanding of contemporary teaching and learning approaches in Embedded Systems and Internet of Things (IoT) technologies. The session focused on hands-on learning methodologies, project-based education, and practical implementation using popular development platforms such as Arduino Uno, ESP32, and Raspberry Pi.

The importance of experiential learning in engineering education. Traditional theory-based learning is gradually being replaced by interactive and hands-on methodologies that improve student engagement and technical competency. The session highlighted the role of open-source hardware and software platforms in facilitating practical learning and rapid prototyping. Demonstrations and examples were provided using Arduino Uno, ESP32, and Raspberry Pi boards.





19/3/26
Course Coordinator

Mrs. R. KEERTHIKA, M.Tech.,

HOD/ECE

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Academic Year: 2025-26 | Even Semester | 6th Semester B.E Programs | Regulation: 2023

2302EC601 – Antennas and Wave Propagation

Module 4 – Horn Antenna – 22.4.2026

Explanation with Lab Components – Horn antenna design using Rectangular Waveguide

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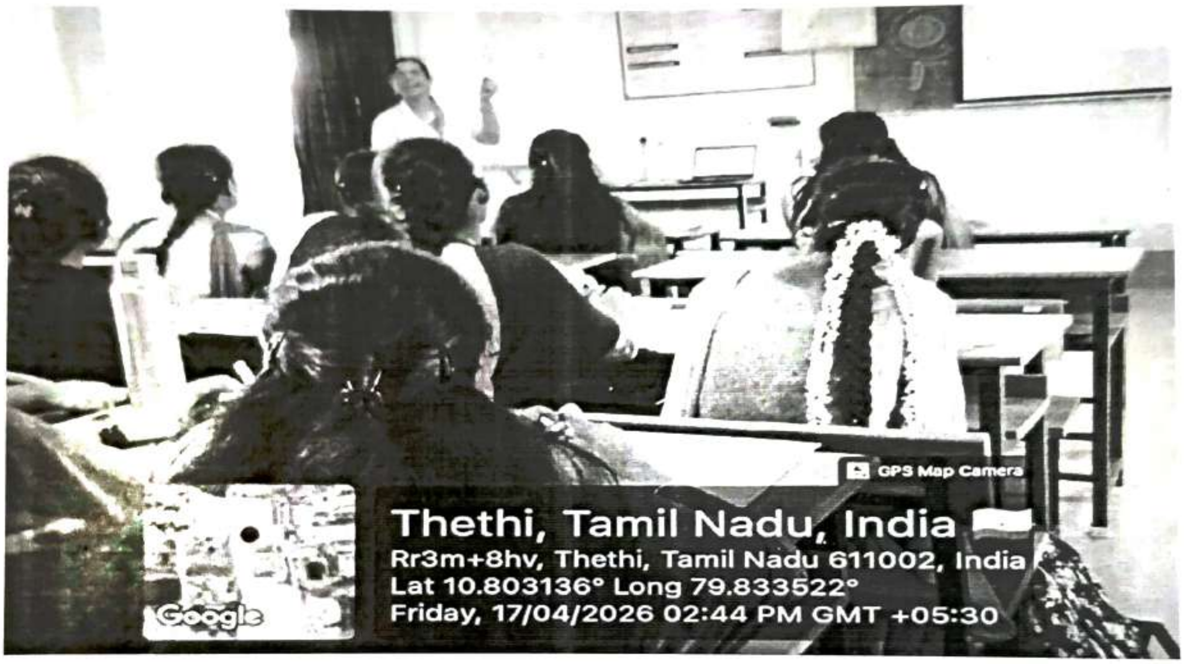
2302EC601 – Antennas and Wave Propagation

Module 4 – Babinet's Principle – 17.4.2026

Real time example - White Paper with slot



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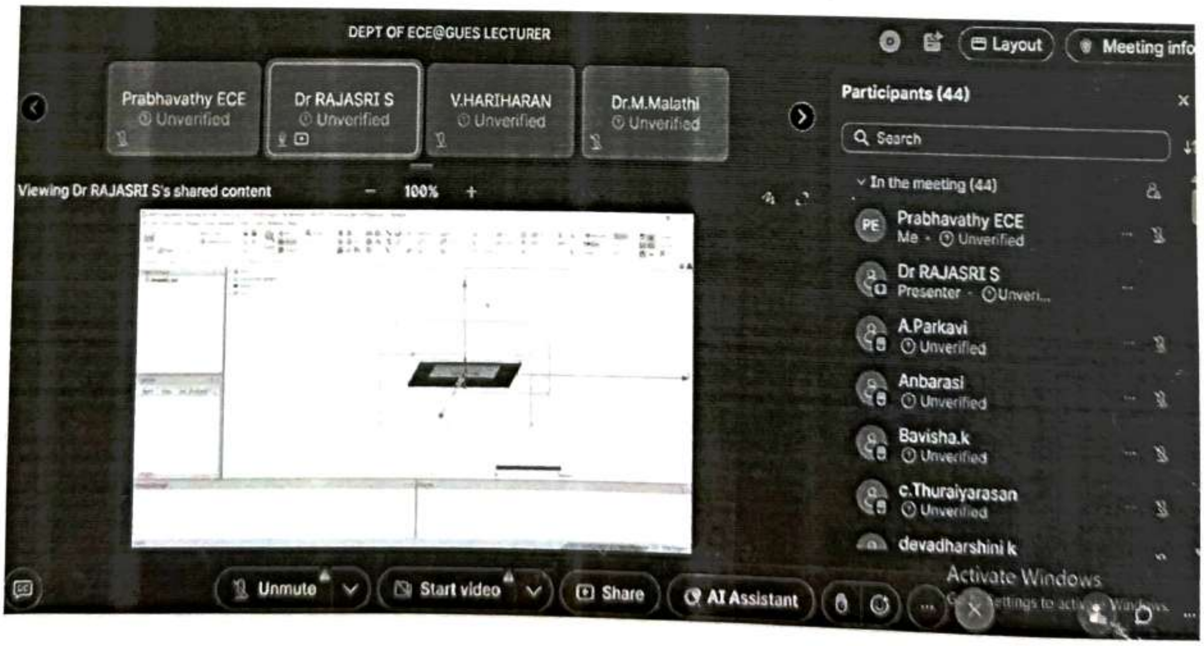
2302EC601 – Antennas and Wave Propagation
ELECTROMAGNETIC SIMULATION OF ANTENNA STRUCTURES USING HFSS
Guest Lecture Webinar on 1.5.2026
Module 3&4



M. M

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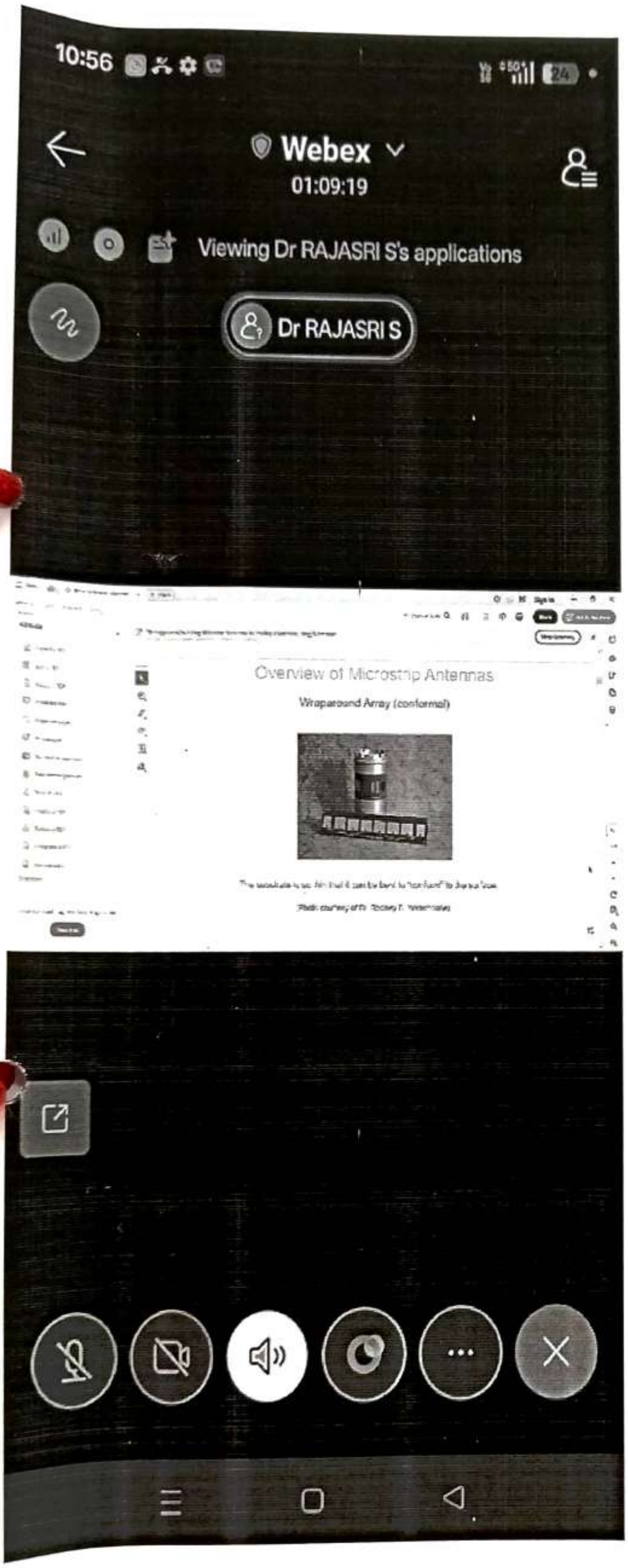

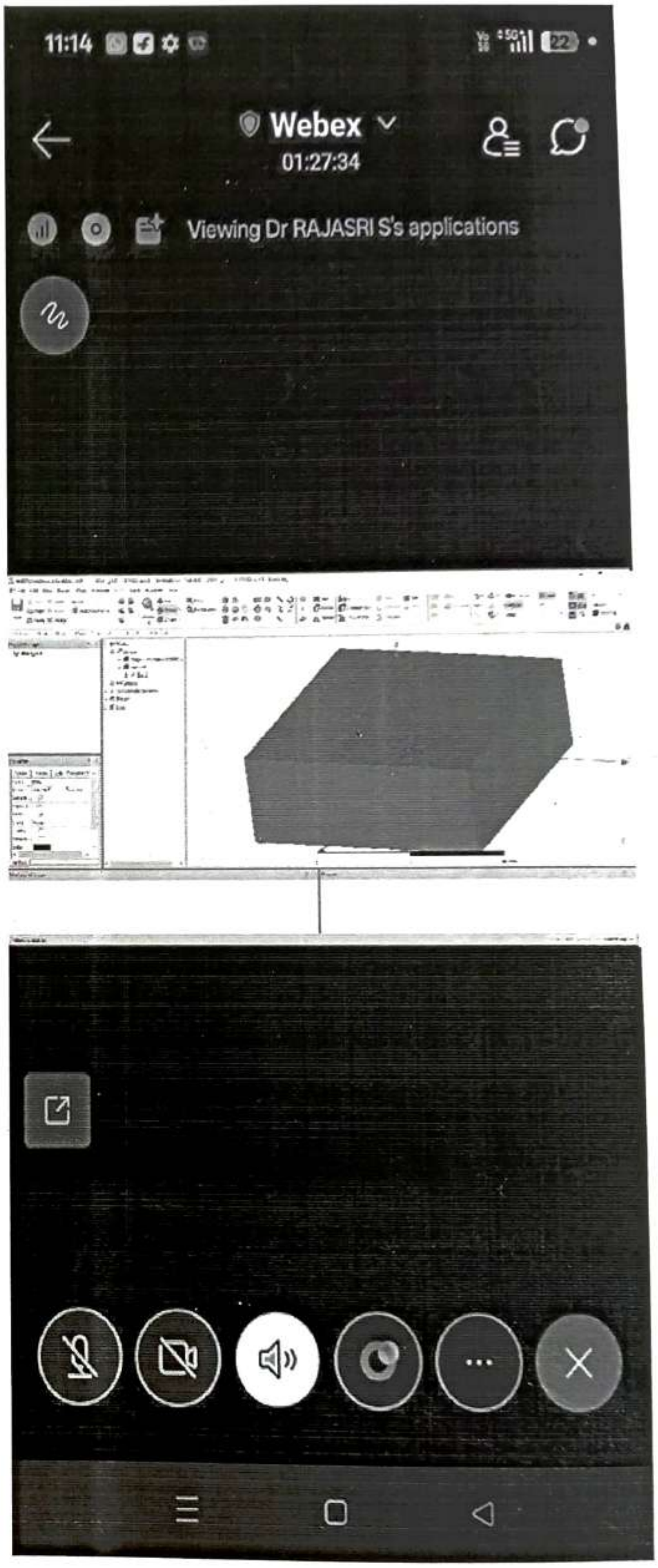

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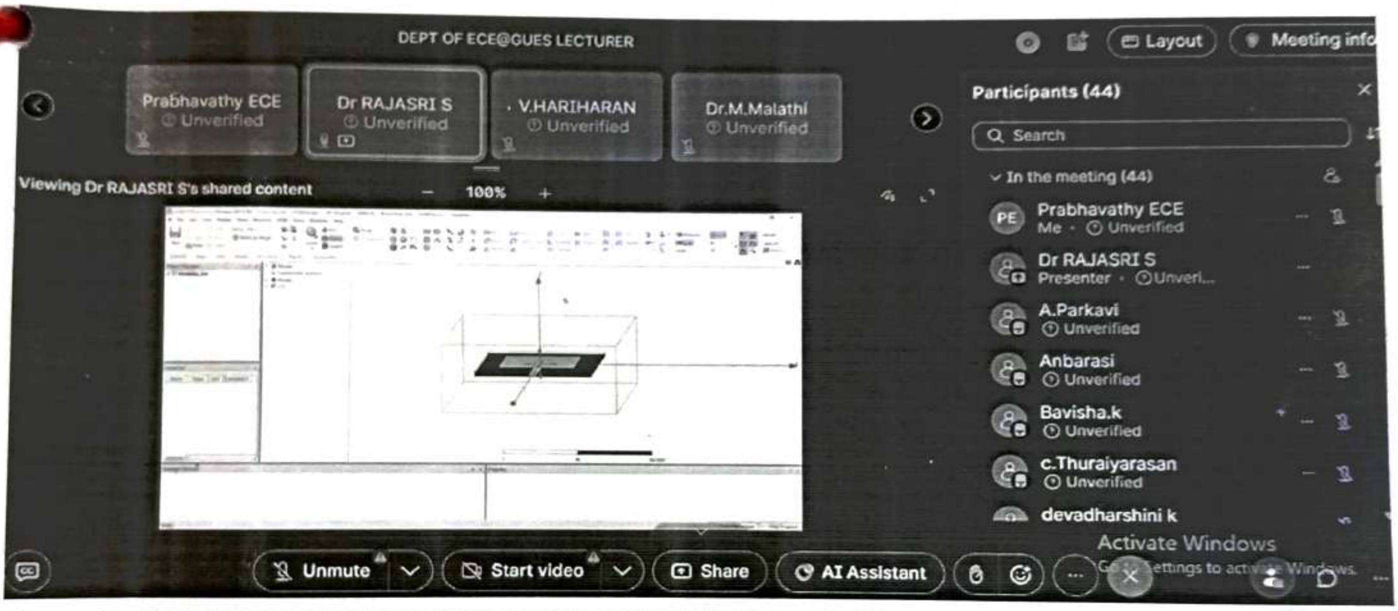

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